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Sprint 4 Individual Contribution Summary

1. Contribution Evidence and Impact

Contribution 1 – Establishing the H1 Humanoid Simulation Environment

Situation

The project required a functioning humanoid simulation environment before any Model Predictive Control (MPC) algorithms could be implemented or evaluated. At the start of Sprint 4, the LearningHumanoidWalking repository had not been fully explored and its architecture was unfamiliar.

Task

My responsibility was to configure the simulation environment, understand the repository structure, and verify that the Unitree H1 humanoid model could successfully execute standing and walking tasks.

Action

I cloned and configured the LearningHumanoidWalking repository, analysed the project architecture, and investigated the environment classes, robot interfaces, task definitions, and observation spaces. I executed multiple validation tests including the standing environment and walking environment. I traced the control flow through the H1 environment implementation to understand how actions, observations, rewards, and termination conditions were handled.

Evidence includes:

- Environment execution logs

H1 standing tests

```
(learninghumanoidwalking) PS C:\Users\rumet\Desktop\MPC test 2\LearningHumanoidWalking\experiments> python test_h1.py
H1 environment started
Observation shape: (43,)
Step:   0 | Height: 0.988 | Collision: False | Reward: 0.252
Step:   1 | Height: 0.981 | Collision: False | Reward: 0.065
Step:   2 | Height: 0.967 | Collision: False | Reward: 0.424
Step:   3 | Height: 0.949 | Collision: False | Reward: 0.408
Step:   4 | Height: 0.928 | Collision: False | Reward: 0.393
Step:   5 | Height: 0.904 | Collision: False | Reward: 0.399
Step:   6 | Height: 0.879 | Collision: False | Reward: 0.370
Step:   7 | Height: 0.852 | Collision: False | Reward: 0.328
Step:   8 | Height: 0.825 | Collision: False | Reward: 0.356
Step:   9 | Height: 0.797 | Collision: False | Reward: 0.348
Step:  10 | Height: 0.769 | Collision: False | Reward: 0.331
Step:  11 | Height: 0.741 | Collision: False | Reward: 0.315
Step:  12 | Height: 0.712 | Collision: False | Reward: 0.307
Step:  13 | Height: 0.684 | Collision: False | Reward: 0.294
Step:  14 | Height: 0.656 | Collision: False | Reward: 0.284
Step:  15 | Height: 0.629 | Collision: False | Reward: 0.276
Step:  16 | Height: 0.603 | Collision: False | Reward: 0.270
Step:  17 | Height: 0.577 | Collision: False | Reward: 0.265

========== TERMINATED ==========
REASON: H1 fell below 0.6 m
==============================
```

Figure 1: H1 walking tests

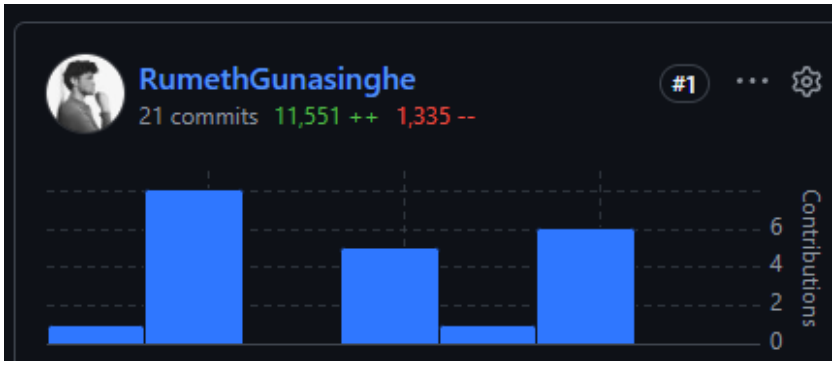


Figure 2: Git commits related to simulation setup

Result

The H1 humanoid simulation environment was successfully validated. I confirmed that the robot could be instantiated, observations could be collected, and rewards could be calculated correctly. This provided the technical foundation required for all subsequent MPC research and implementation activities.

Time spent: 10 hours

Contribution 2 – Joint Influence Analysis and System Identification

Situation

Before designing an MPC controller, it was necessary to understand how different lower-body joints influenced humanoid balance and body orientation.

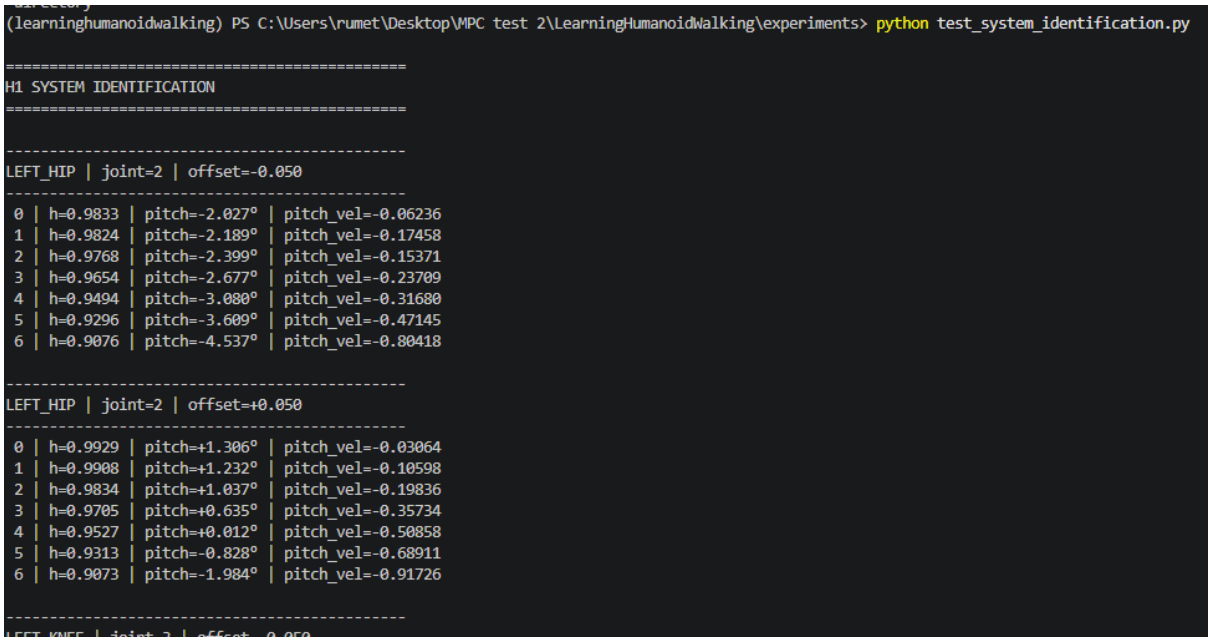
Task

I was responsible for performing system identification experiments and analysing the effect of individual joints on robot dynamics.

Action

I developed and executed joint-direction testing experiments and system-identification tests. These experiments systematically perturbed hip and knee joints while recording pelvis height, pitch angle, and pitch velocity. The resulting data was analysed to determine which joints had the greatest influence on balance recovery and posture control.

Evidence includes:



Result

The experiments successfully identified key joints affecting sagittal-plane stability and provided an initial understanding of robot dynamic behaviour. These findings informed future MPC controller design and reduced uncertainty regarding actuator influence.

Time spent: 8 hours

Contribution 3 – Development of Initial MPC Prototype

Situation

The project aims to implement Whole-Body Model Predictive Control. However, implementing full humanoid MPC directly would introduce excessive complexity and risk.

Task

I was responsible for creating an initial reduced-order MPC proof-of-concept.

Action

I researched Linear Inverted Pendulum Models (LIPM), reviewed MPC literature, and implemented a simplified predictive controller to regulate centre-of-mass motion. I developed experiments to evaluate controller performance and analysed the resulting trajectories.\

Evidence includes:

```
(learninghumanoidwalking) PS C:\Users\rumet\Desktop\MPC test 2\LearningHumanoidWalking\experiments> python test_mpc.py
Initial COM velocity : 0.000 m/s

Step 00 | COM: +0.09995 m | Velocity: -0.00376 m/s | ZMP: +0.11500 m
Step 01 | COM: +0.09981 m | Velocity: -0.00754 m/s | ZMP: +0.11500 m
Step 02 | COM: +0.09958 m | Velocity: -0.01137 m/s | ZMP: +0.11500 m
Step 03 | COM: +0.09926 m | Velocity: -0.01401 m/s | ZMP: +0.11000 m
Step 04 | COM: +0.09887 m | Velocity: -0.01675 m/s | ZMP: +0.11000 m
Step 05 | COM: +0.09842 m | Velocity: -0.01959 m/s | ZMP: +0.11000 m
Step 06 | COM: +0.09791 m | Velocity: -0.02130 m/s | ZMP: +0.10500 m
Step 07 | COM: +0.09735 m | Velocity: -0.02314 m/s | ZMP: +0.10500 m
Step 08 | COM: +0.09675 m | Velocity: -0.02513 m/s | ZMP: +0.10500 m
Step 09 | COM: +0.09611 m | Velocity: -0.02602 m/s | ZMP: +0.10000 m
Step 10 | COM: +0.09545 m | Velocity: -0.02708 m/s | ZMP: +0.10000 m
Step 11 | COM: +0.09476 m | Velocity: -0.02830 m/s | ZMP: +0.10000 m
Step 12 | COM: +0.09403 m | Velocity: -0.02970 m/s | ZMP: +0.10000 m
Step 13 | COM: +0.09329 m | Velocity: -0.03004 m/s | ZMP: +0.09500 m
Step 14 | COM: +0.09253 m | Velocity: -0.03056 m/s | ZMP: +0.09500 m
Step 15 | COM: +0.09176 m | Velocity: -0.03128 m/s | ZMP: +0.09500 m
Step 16 | COM: +0.09096 m | Velocity: -0.03219 m/s | ZMP: +0.09500 m
Step 17 | COM: +0.09016 m | Velocity: -0.03205 m/s | ZMP: +0.09000 m
Step 18 | COM: +0.08936 m | Velocity: -0.03211 m/s | ZMP: +0.09000 m
Step 19 | COM: +0.08855 m | Velocity: -0.03237 m/s | ZMP: +0.09000 m
Step 20 | COM: +0.08774 m | Velocity: -0.03283 m/s | ZMP: +0.09000 m
Step 21 | COM: +0.08691 m | Velocity: -0.03350 m/s | ZMP: +0.09000 m
Step 22 | COM: +0.08608 m | Velocity: -0.03313 m/s | ZMP: +0.08500 m
Step 23 | COM: +0.08525 m | Velocity: -0.03296 m/s | ZMP: +0.08500 m
Step 24 | COM: +0.08443 m | Velocity: -0.03300 m/s | ZMP: +0.08500 m
Step 25 | COM: +0.08360 m | Velocity: -0.03325 m/s | ZMP: +0.08500 m
Step 26 | COM: +0.08276 m | Velocity: -0.03370 m/s | ZMP: +0.08500 m
Step 27 | COM: +0.08193 m | Velocity: -0.03312 m/s | ZMP: +0.08000 m
Step 28 | COM: +0.08111 m | Velocity: -0.03274 m/s | ZMP: +0.08000 m
Step 29 | COM: +0.08029 m | Velocity: -0.03256 m/s | ZMP: +0.08000 m
Step 30 | COM: +0.07948 m | Velocity: -0.03259 m/s | ZMP: +0.08000 m
Step 31 | COM: +0.07866 m | Velocity: -0.03282 m/s | ZMP: +0.08000 m
Step 32 | COM: +0.07785 m | Velocity: -0.03201 m/s | ZMP: +0.07500 m
Step 33 | COM: +0.07706 m | Velocity: -0.03140 m/s | ZMP: +0.07500 m
Step 34 | COM: +0.07628 m | Velocity: -0.03098 m/s | ZMP: +0.07500 m
Step 35 | COM: +0.07551 m | Velocity: -0.03076 m/s | ZMP: +0.07500 m
Step 36 | COM: +0.07474 m | Velocity: -0.03073 m/s | ZMP: +0.07500 m
Step 37 | COM: +0.07397 m | Velocity: -0.03089 m/s | ZMP: +0.07500 m
Step 38 | COM: +0.07319 m | Velocity: -0.03124 m/s | ZMP: +0.07500 m
Step 39 | COM: +0.07242 m | Velocity: -0.03054 m/s | ZMP: +0.07000 m
```

Result

The prototype successfully demonstrated predictive regulation of centre-of-mass motion and provided evidence that MPC can be integrated into the humanoid control framework. This significantly reduced technical risk for future sprints.

Time spent: 12 hours

Total Time Spent

30 hours

2. Teamwork, Communication and Professional Practice

Throughout Sprint 4, I maintained regular communication with my teammate, supervisor, and client. I attended team discussions, supervision meetings, and client meetings and provided updates on technical progress. When communication delays

occurred earlier in the project, I proactively followed up with stakeholders and ensured the team remained informed.

I collaborated closely with Thinidu while conducting research and implementation tasks. We regularly reviewed findings together and discussed technical decisions regarding simulation design and MPC architecture.

3. Reflection, Learning and Continuous Improvement

Sprint 4 significantly improved my understanding of humanoid robotics and predictive control. One challenge was the complexity of the LearningHumanoidWalking framework, which required substantial time to understand before meaningful development could begin.

I learned that validating assumptions through experimentation is essential before implementing advanced control algorithms. The system-identification experiments revealed behaviour that was not immediately apparent from reviewing source code alone.

In future sprints, I plan to focus on integrating MPC directly into the H1 simulation and developing quantitative evaluation metrics. I will also allocate more time to controller validation before attempting more advanced locomotion tasks.